Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: Building Customer analysis Dashboard

Submitted By:

Aarush C S

Reg. No.: 2411021240028

Date: 16/10/2026

Submitted To:

Faculty Name: Mr. Devatha Nikhil

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

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The purpose of this experiment is to build a Customer Analysis Dashboard in Microsoft Power BI to examine customer purchasing behaviour, product-category activity, payment preferences, and revenue trends.

1. Problem Statement

Businesses need to understand customer purchasing behaviour and revenue performance to make informed decisions. By analysing transaction data, a business can identify popular product categories, preferred payment methods, monthly revenue patterns, frequent purchasers, and the extent to which revenue depends on a small group of customers.

2. Objectives

1. **Identify the product category with the highest purchasing activity.**
2. **Determine the most frequently used payment method.**
3. **Analyse how revenue changes from month to month.**
4. **Identify customers who make the most purchases.**
5. **Assess whether a small group of customers contributes a large proportion of revenue.**
6. **Present key business metrics and insights through Power BI visuals.**

5. Data Import and Preparation

7. Imported the customer and order CSV files into Power BI Desktop.
8. Checked field names and data types, especially date and numeric columns.
9. Prepared the order date for monthly analysis.
10. Created a relationship between customer and order tables using CustomerID.
11. Created measures for total revenue, total orders, customer counts, and average order value.
12. Built visuals to compare monthly revenue, product categories, payment methods, customer purchase counts, and customer revenue.

6. Key Measures

The following measures are used or can be used for the dashboard. Adjust table names if your model uses different names.

Measure	DAX formula
Total Revenue	Total Revenue = SUM('customer_analysis_orders_5000'[SalesAmount])
Total Orders	Total Orders = DISTINCTCOUNT('customer_analysis_orders_5000'[OrderID])

Purchasing Customers	Purchasing Customers = DISTINCTCOUNT('customer_analysis_orders_5000'[CustomerID])
Average Order Value	Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
Customer Purchase Count	Customer Purchase Count = DISTINCTCOUNT('customer_analysis_orders_5000'[OrderID])

7. Dashboard Screenshots

Insert the screenshots of the completed Power BI dashboard in the spaces below. Keep the screenshots clear and readable.

7.1 Overall Customer Analysis Dashboard

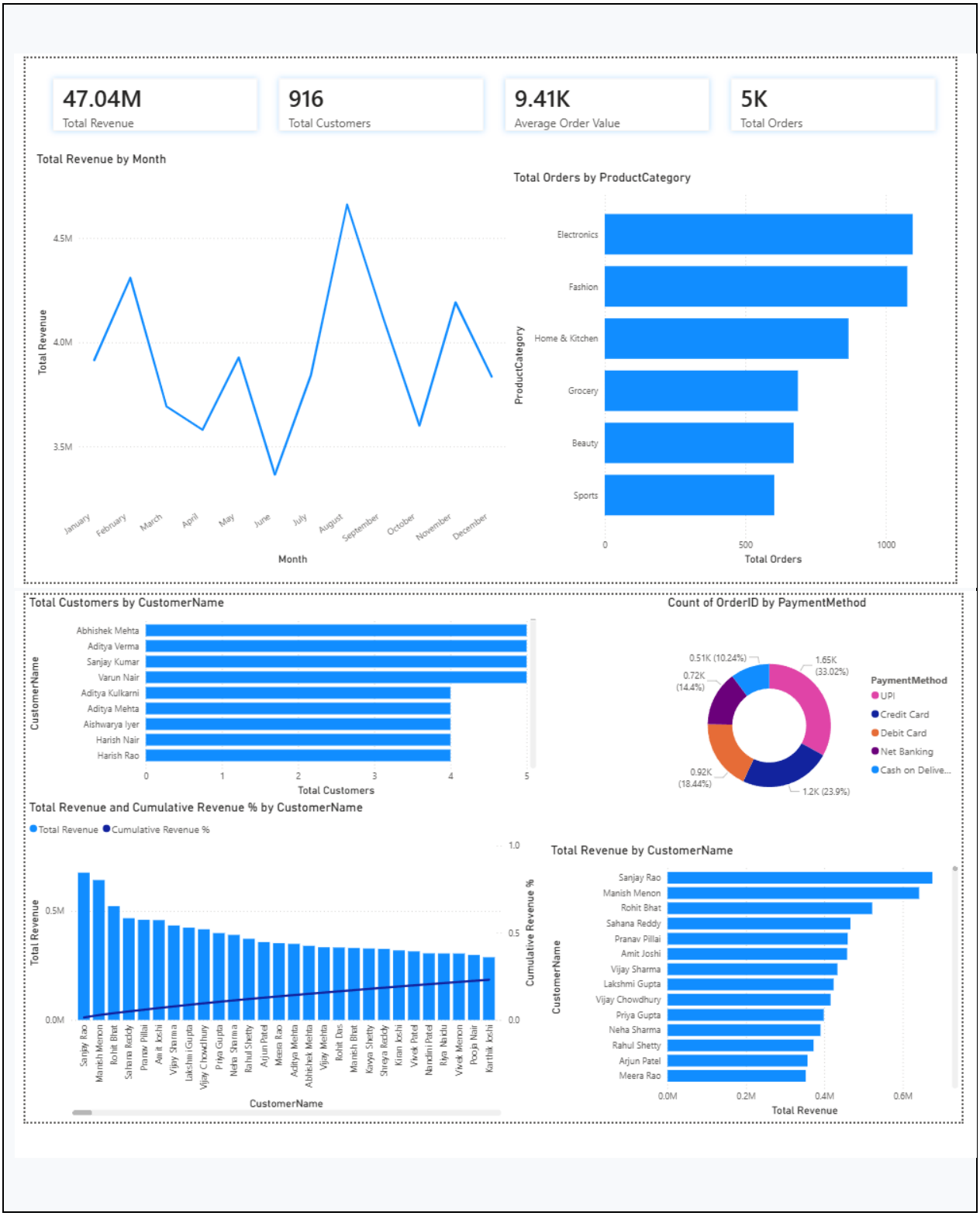


Figure: 7.1 Overall Customer Analysis Dashboard.

7.2 Monthly Revenue and Product Category Analysis

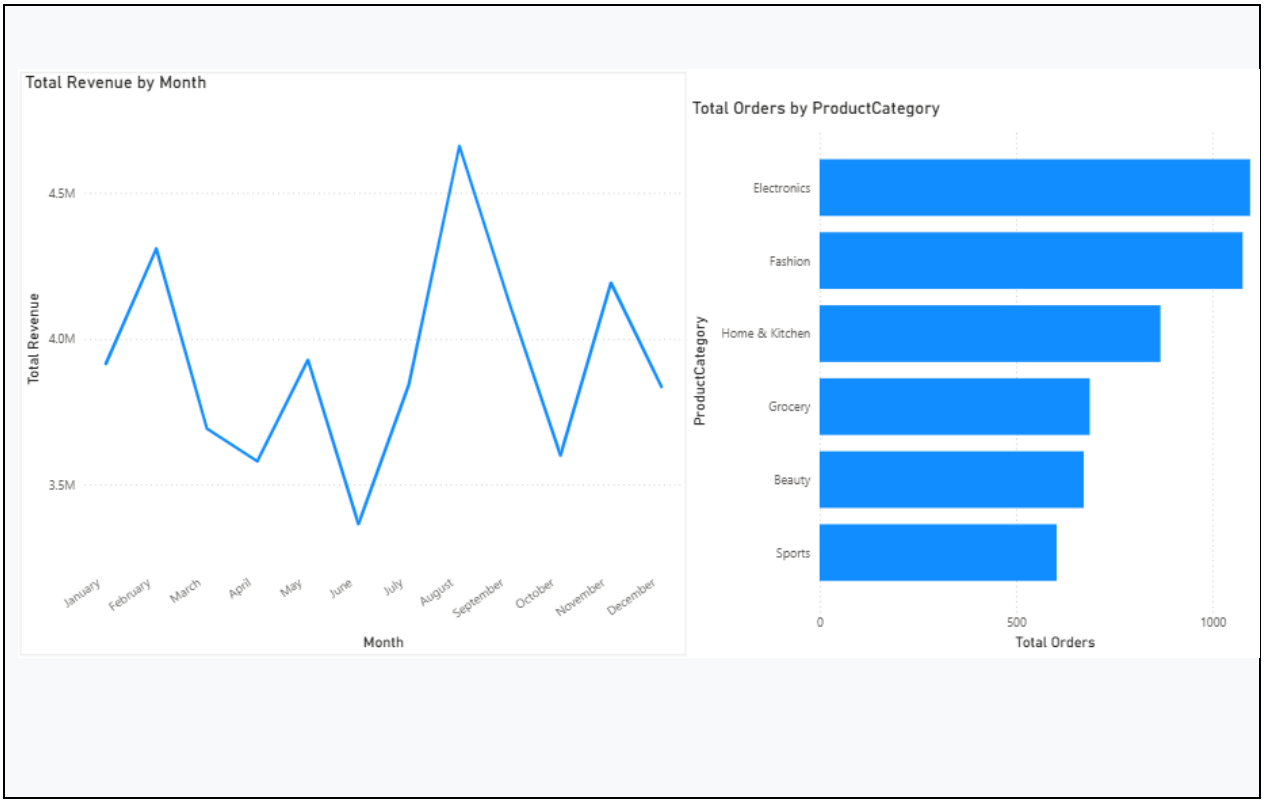


Figure: 7.2 Monthly Revenue and Product Category Analysis.

7.3 Payment Method and Customer Purchase Analysis

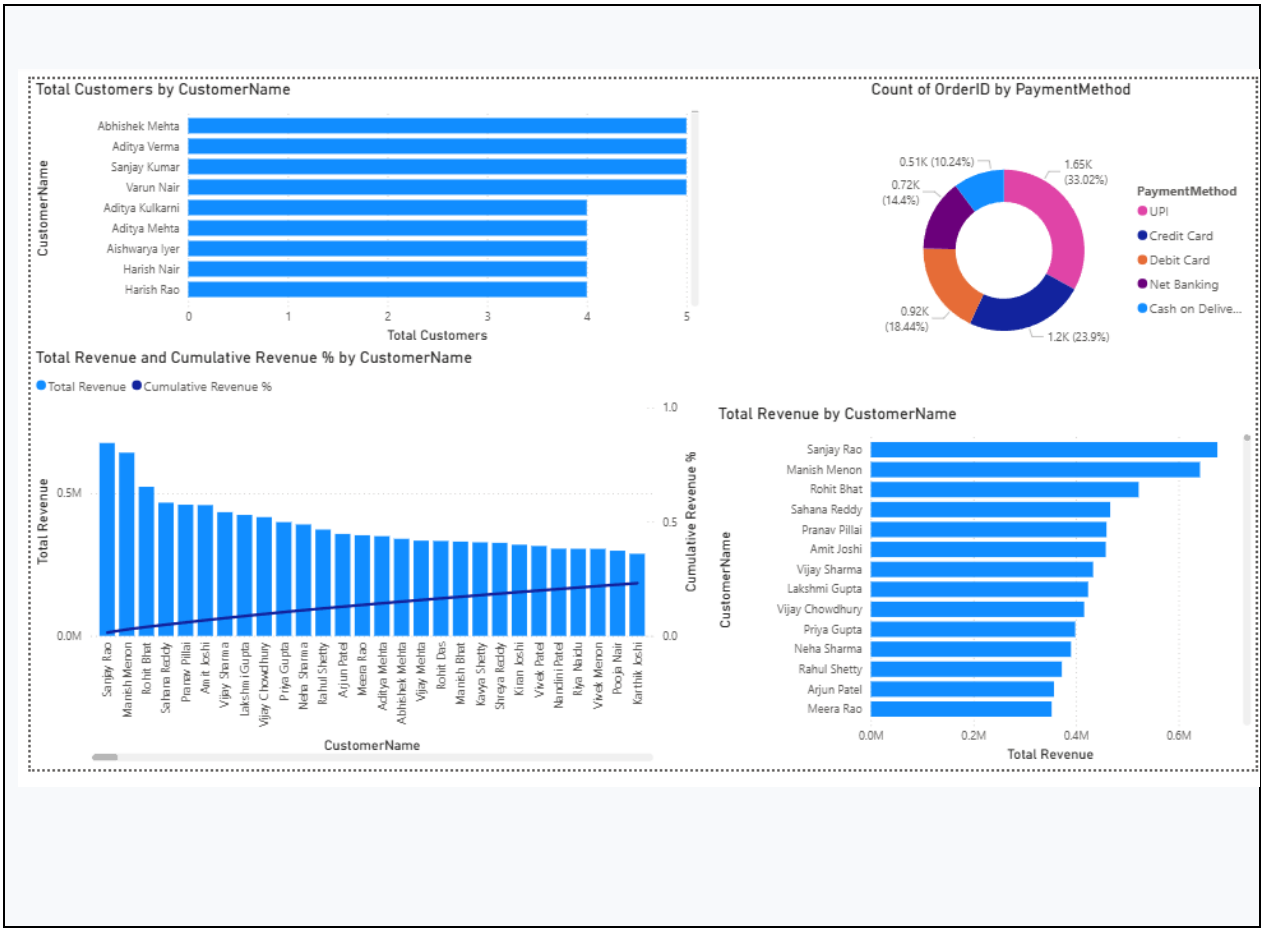


Figure: 7.3 Payment Method and Customer Purchase Analysis.

7.4 Customer Revenue Concentration Analysis

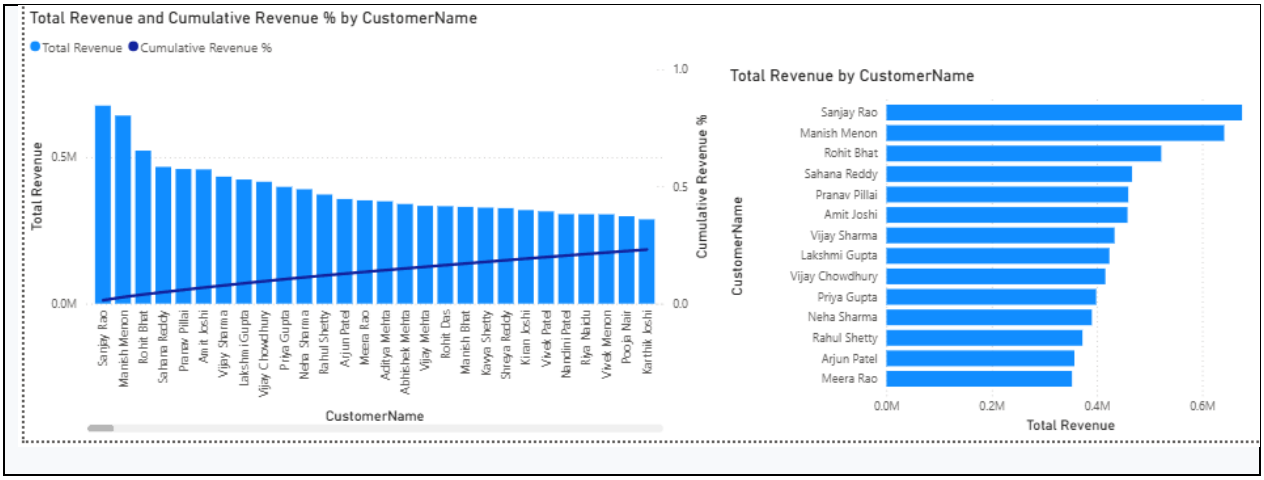


Figure: 7.4 Customer Revenue Concentration Analysis.

8. Analysis and Answers to Business Questions

8.1 Which product category is most popular?

A horizontal bar chart comparing orders by ProductCategory is used to identify the category with the greatest purchasing activity. In the provided dashboard screenshot, Electronics has the longest bar, followed closely by Fashion. Therefore, Electronics appears to be the most popular category by the displayed order count. This conclusion refers to order activity, not necessarily the highest profit.

8.2 Which payment method is most frequently used?

A donut chart compares the count of OrderID by PaymentMethod. In the provided screenshot, UPI has the largest segment at approximately 1.65K orders (33.02%), followed by Credit Card at approximately 1.2K (23.9%) and Debit Card at approximately 0.92K (18.44%). Thus, UPI is the most frequently used payment method in the displayed results.

8.3 How does revenue change month by month?

A line chart titled 'Total Revenue by Month' is used to compare revenue over the months. In the provided screenshot, revenue rises from January to February, falls through March and April, recovers in May, and reaches a low point in June. It rises in July and reaches its highest displayed point in August, then falls in September and October, rises in November, and declines in December. The chart should be checked to ensure the month sequence is chronologically sorted and that the date field has no blank categories.

8.4 Which customers have made the most purchases?

A horizontal bar chart showing customer purchase counts is used to identify frequent purchasers. In the provided screenshot, several customers have five orders, including Abhishek Mehta, Aditya Verma, and Sanjay Kumar. Several other displayed customers have four orders. The chart should be sorted by purchase count in descending order, and a Top N filter can be applied to show the ten highest customers.

8.5 Are a small number of customers responsible for a large proportion of revenue?

A customer revenue bar chart and a cumulative revenue percentage line can be used to evaluate revenue concentration. The screenshot shows revenue values for individual customers, but the cumulative percentage line rises gradually and the chart does not clearly establish the share contributed by the top 10% or 20% of customers. Therefore, a firm conclusion cannot be made from the screenshot alone. To answer this question accurately, calculate the cumulative revenue percentage across all customers and report the share contributed by the top 10% or 20%.

9. Key Business Insights

- Total revenue shown on the dashboard is approximately 47.04M.
- The dashboard shows approximately 5K total orders.
- The customer KPI displays 916 customers.
- Average order value is displayed as approximately 9.41K.
- Electronics appears to have the highest order count among the displayed product categories.
- UPI is the most frequently used payment method in the displayed results.
- The monthly revenue chart shows noticeable variation, with August as the highest displayed month and June as the lowest displayed month.
- The highest displayed customer purchase count is five orders.
- Customer revenue concentration requires a cumulative percentage calculation before making a definite claim about dependence on a small group of customers.

10. Limitations

The dataset fields visible in the reference dashboard include sales amount but do not show a product cost or profit field. Consequently, actual profit and profit margin cannot be calculated reliably unless cost data is added. The customer concentration conclusion also requires a correctly calculated cumulative revenue percentage across all customers. Finally, the monthly chart should be checked for chronological sorting and missing date values.

11. Conclusion

The Customer Analysis Dashboard brings together KPI cards, a monthly revenue line chart, category and customer bar charts, and a payment-method donut chart. The visuals help identify product categories with strong purchasing activity, understand payment preferences, compare monthly revenue, and find frequent purchasers. A cumulative revenue percentage measure is required to assess whether

revenue is concentrated among a small group of customers. The dashboard demonstrates how Power BI can convert transaction data into business insights that support data-driven decisions.